

Sequences and Series



Sequence limits

- 1 Determine whether each sequence converges, and find the limit if it does:

a $a_n = \frac{3n+1}{n+2}$

b $a_n = \frac{n^2}{2^n}$

c $a_n = (-1)^n$

- 2 Evaluate $\lim_{n \rightarrow \infty} \left(1 + \frac{3}{n}\right)^n$.
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Series and partial sums

- 3 Determine whether $\sum_{n=1}^{\infty} \frac{1}{n}$ converges or diverges, and name the test or fact that justifies your answer.

- 4 Evaluate the geometric series $\sum_{n=0}^{\infty} 3\left(\frac{2}{5}\right)^n$.

- 5 Use partial fractions and telescoping to evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.
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Monotonic and bounded sequences

- 6 Show that the sequence $a_n = \frac{n}{n+1}$ is increasing, by comparing a_{n+1} and a_n .
- 7 Explain why the sequence $a_n = \frac{n}{n+1}$ (increasing and bounded above by 1) must converge, and find its limit.
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More series practice

- 8 Evaluate the series $\sum_{n=1}^{\infty} \left[\left(\frac{1}{2}\right)^n + \left(\frac{1}{3}\right)^n \right]$ by splitting into two geometric series.
- 9 A ball dropped from 10 m rebounds to 60% of its previous height each bounce. Using an infinite geometric series, find the total distance traveled.
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Recursive sequences

- 10 A sequence is defined recursively by $a_1 = 2$, $a_{n+1} = \frac{a_n + 6}{2}$. Compute a_2 , a_3 , and a_4 .

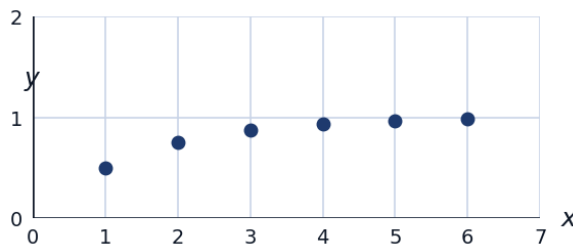
- 11 Assuming the recursive sequence above converges to a limit L , find L by setting $L = \frac{L+6}{2}$ and solving.
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Repeating decimals as series

- 12 Express the repeating decimal $0.\overline{45}$ as a fraction using an infinite geometric series with $a = \frac{45}{100}$ and $r = \frac{1}{100}$.
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Visualizing partial sums of a series

- 13 The graph shows the first six partial sums $S_n = \sum_{k=1}^n \left(\frac{1}{2}\right)^k$ of a geometric series. Use the pattern to conjecture the value the sums are approaching, and confirm it using the geometric series formula.



Series with factorial and combined terms

- 14 Evaluate $\sum_{n=0}^{\infty} \frac{1}{2^n} - \sum_{n=0}^{\infty} \frac{1}{3^n}$ by evaluating each geometric series separately.
- 15 A sequence is defined by $a_n = \frac{2n + (-1)^n}{n}$. Find $\lim_{n \rightarrow \infty} a_n$, explaining why the $(-1)^n$ term does not affect the limit.
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The Squeeze Theorem for sequences

- 16 Use the squeeze theorem to find $\lim_{n \rightarrow \infty} \frac{\sin n}{n}$, using the fact that $-1 \leq \sin n \leq 1$ for all n .
- 17 Use the squeeze theorem to find $\lim_{n \rightarrow \infty} \frac{n!}{n^n}$, using the bound $0 < \frac{n!}{n^n} \leq \frac{1}{n}$ for $n \geq 1$.
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Applications of infinite series

- 18** A patient takes a 100 mg dose of medication every 8 hours. Between doses, 75% of the drug remaining is eliminated, so a fraction $r = 0.25$ carries over to just before the next dose. Using an infinite geometric series, find the long-run steady-state amount in the body immediately after a dose (assume doses accumulate indefinitely).